

Hemant Kumar

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Professional Summary

Software Engineer with 7+ years of experience building distributed systems in AdTech and cloud-native environments. Expertise in Scala, Golang, Apache Spark, Akka, AWS, and Databricks. Proven success in delivering scalable solutions, automating pipelines, and optimizing high-throughput services.

Experience

Senior Software Engineer II, Alphonso Labs (LG Ad Solutions), Bangalore | July 2019 – Present

- **DSP Data Augmenter** | *Golang, go-fiber, Redis, Spark, AWS EC2, ELB, Microservices, Kubernetes*
 - Built a high-performance bid request augmenter service, replacing third-party solutions
 - Built Golang based API server using go-fiber with Redis as hot store for segments
 - Developed a Spark application to ingest/evict segments into/from Redis cluster
 - Achieved **45k QPS** with p99 latency of **10ms** on EC2 r5.2xlarge instances
 - Reduced segment activation turnaround time from nearly 2 days to **30 minutes**
 - Developed monitoring and alerting tools for Redis and database consistency
- **Audience Builder** | *ReactJS, Apache Spark, Akka, REST, AWS, Data Pipelines*
 - Designed and developed a scalable audience generation platform used for ad targeting
 - Implemented a DAG-based UI for defining targeting criteria using filter, aggregation, and join nodes
 - Built REST APIs with Akka-HTTP and developed a React-based self-serve UI
 - Utilized Akka-Actor for job lifecycle management and Spark for querying and generating audience data
 - Enabled multi-protocol data delivery to various DSPs and S3
 - Automated workflows to scale daily audience generation jobs from 30 to **400**, reducing human errors
- **AWS & Databricks Migration** | *AWS, EKS, ElastiCache, RDS, Databricks*
 - Led migration of Audience Builder and DSP Data Augmenter from on-prem Spark to AWS and Databricks
 - Authored Helm charts for deploying services on Amazon EKS
 - Built an SBT plugin for automated JAR packaging and deployment to Databricks
 - Migrated persistence and caching layers to Amazon RDS and ElastiCache
- **Attribution Reporting Framework** | *Spring Boot, Apache Camel, Kubernetes, Redis, Microservices, Event Driven*
 - Enabled advertisers to measure ad impact with reach, conversion, and lift metrics
 - Developed Spring Boot microservices and used Apache Camel for inter-service communication
 - Used Redis as the task queue for workers and persistent state layer
 - Reduced manual effort of monitoring external data for reporting by **100%**

Software Engineer, Cavium Networks (now Marvell), Hyderabad | July 2015 – June 2016

- **ZIP Compression Accelerator** | *Kernel Drivers, DLKM, ZLIB, LZ*
 - Developed Linux kernel drivers for ThunderX ZIP compression hardware
 - Created user-space libraries overriding libz.so to utilize hardware accelerators
 - Contributed code to the Linux kernel under `/drivers/crypto/cavium/zip`

Education

Master of Technology, Computer Science and Engineering, IIT Kanpur, India (2019), CGPA: 8.67

Bachelor of Technology, Computer Science and Engineering, NIT Calicut, India (2015), CGPA: 8.52

Technical Skills

Languages Scala, Golang, JavaScript, C, C++, HTML, CSS, Python, Bash, ~~TeX~~

Frameworks/Libraries Spring Boot, Apache Camel, Akka, Spark, ReactJS, Redux

Databases/Caches Redis, MySQL, Hazelcast

Cloud/Tools AWS, Jenkins, Rundeck, Airflow, Docker, Kubernetes, Databricks, SBT

Academic Projects

- **Base-3 Normal Number Construction in ACO**, under Dr. Satyadev Nandakumar, IIT Kanpur
 - Explored the possibility of having non-uniform ACO circuit for querying the approximate base-2 number

Achievements

- 1st Dan Black Belt in Okinawan Shorin-ryu Karate and Kukkiwon Taekwondo